

Machine learning and artificial intelligence is not suited for every application; therefore, it is important to understand when these tools can be applied to benefit the organization. For a financial services firm, there are a few concepts that are critical to developing artificial intelligence-based tools and services that are offered to the end consumers. The five concepts critical for building ethical AI applications in the accounting and finance sector are: unbiased, diversity, privacy and data governance, laws and regulations, and performance.

Let's begin by talking about data, the foundation of artificial intelligence and machine learning. The importance of data acquisition and preparation is undervalued in the development of AI systems. If there is a corruption in preparing the data, or the acquired data is not authentic, these flaws are known as data risks. The data being acquired is defined by the problem at hand. Acquisition of data and the suitability of data must match the problem. More importantly, the acquired data must cover all instances and possibilities of the inputs to the algorithm. For example, Apple Card's lending algorithms gave more credit to husbands compared to their wives, even if the wife had a better credit score and income. Facebook's ad platform targets its audience for employment and housing ads based on race and gender (Frank and Ke). The root cause of these flaws in AI systems is biased datasets. When the data acquired does not suit the problem identified or the data is not diverse enough, it is known as design risk. The relationship between design risk and data risk is very well explained in the 'EY Ethical Framework: 5 attributes necessary to sustain in AI' (Cobey and Boillet). When applications are applied at large scale like for insurance companies and wealth management, the algorithm should be trained to deal with all kinds of consumers, and this can be done by expanding the data being trained on and adding all kinds of scenarios of the data that are possible. In simple terms, this refers to diversity in the dataset.

How can the development team behind an AI project make sure that the data set contains diverse records and instances of the product they are building? This question is very hard to answer since it is almost impossible to ensure that the records contain all possibilities and diverse factors for training data. However, including a diverse set of individuals when building the application is one way to increase the chances of having many diverse ideas included in the project and hence one way of making sure that the dataset contains all instances of real-life scenarios so that the training incorporates these diverse scenarios and the model created in the end is unbiased. The European Union Trustworthy AI Assessment lists diversity, non-discrimination, and fairness as one of their assessment guidelines in their ethical framework for development of artificial intelligence (Lemonne).

The definition of data risks is very narrow when looking at the EY ethical framework. Data risks should also include the possibility of losing the data, the data getting stolen, data leaks and other ways private information is made public. This means that the privacy and governance of data is also a risk that a firm should consider when building AI systems. The firm should look at these as organizational risks because they can cause severe harm to the company itself. If data is lost because of hackers or if the user data is intentionally sold to other companies without the consent of the users, it ultimately reflects very badly on the firm and the reputation of the company is tarnished. Therefore, the organization should view data privacy and governance as a duty rather than an obligation. In the previously mentioned ethical framework provided by the European Union, privacy and data governance has been listed as a pillar of ethical AI assessment. Privacy governance rules should be flexible, and the users should be allowed to have access to their data. The company must also have the utmost respect for the privacy and data ownership of the data owned by their users. Users must be notified that their data is being used for something new like a project or to build a tool.

The application being developed must be useful in the end and there should be a certain level of interpretability that can be extracted out of the model or visualizations. PwC's Responsible AI Toolkit includes interpretability and explain ability in its ethical framework (PricewaterhouseCoopers). This part of the ethical framework relates to the performance of the model as a whole and the interpretation of the model. For example, if the insurance ratings are computed for thousands of consumers but the range of the ratings is too narrow, it does not provide much context from the model and therefore has low explain ability. This means that the data might be clean and authentic, but the model applied might not be appropriate even if the results seem sound. This means that the firm should re-evaluate the model and improve upon it to make sure that the deployed model in the end performs to the promised standards.

PwC also lists regulations as one of their key elements in the ethical frameworks. This can be modified to include laws and regulations related to the AI system being developed. It is important to make sure that the firm developing such instruments is up to date with the laws in the industry and they can factor in these laws and regulations as they are updated into the algorithms that for example, score individuals for internal business purposes or suggest and recommend consumers with financial advice. If the AI system is not in line with the laws, regulations, and ethical standards of society, it will not be able to provide the best social impact possible. At the foundation, every AI application should consider the fundamental rights, human rights, privacy rights, and basic moral values that govern society, and the algorithms should consider these factors as well.

Citations

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